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GenAI- Lab 6
Class Diagram

UML Class Diagram Generation using Local LLM - Experimental Report

1. Base Research Paper

This work is based on the research paper titled:

“Evaluating Large Language Models in Exercises of UML Class Diagram Modeling” (ESEM 2024) by Daniele De Bari et al.

The paper evaluates the capability of Large Language Models (LLMs) to generate UML class diagrams from natural language requirements and compares them with human-generated diagrams.

The evaluation is based on three quality dimensions:
• Syntactic Quality
• Semantic Quality
• Pragmatic Quality

It also compares generated diagrams with reference solutions using semantic distance.

2. Methodology Followed in This Work

The methodology implemented in this project closely follows the experimental setup described in the base paper. The system takes natural language requirements as input and generates UML class diagrams using a local LLM.

The pipeline consists of:
1. Input: Natural language transcript
2. LLM generates PlantUML code
3. PlantUML renders class diagram
4. Evaluation based on defined metrics

3. Prompt Used

The following prompt is used for UML generation:

"Create a UML Class Diagram of the given exercise and give me the PlantUML code."

This prompt is taken directly from the base research paper to replicate baseline conditions.

It ensures:
• Clear task definition (UML Class Diagram generation)
• Context awareness (exercise description)
• Structured output (PlantUML code)

A minimal prompt is intentionally used to evaluate the raw capability of the LLM.

4. Exercises Used for Experimentation

The first four exercises from the provided dataset were used for evaluation:

1. Project Management System
2. Hollywood Approach
3. Word Processor
4. Patient Record and Scheduling System

These exercises cover diverse domains and varying complexity levels, enabling comprehensive evaluation.

5. Evaluation Strategy

The evaluation follows the same quality dimensions defined in the base paper:

Syntactic Quality:
- Checks correctness of UML syntax (e.g., naming, cardinality, symbols)

Semantic Quality:
- Validity and completeness of domain representation

Pragmatic Quality:
- Readability and understandability of diagrams

Additionally, size metrics are computed using code parsing:
- Number of Classes
- Attributes
- Methods
- Relationships

6. Implementation Details

The system is implemented using:
• Streamlit for UI

• Ollama (local LLM) for generation
• PlantUML API for diagram rendering
• Python-based parsing for size metrics
• LLM-based evaluation for quality metrics

7. Experimental Results

exercise	classes	attributes	methods	relationships
1	12	15	10	9
2	6	15	0	0
3	14	25	0	10
4	9	37	0	0

This section summarizes the results obtained from the four exercises.
(You can insert observations, comparisons, and analysis here.)

8. Experimental Outputs (Diagrams & Screenshots)

Insert screenshots of:
• Generated UML code
• Rendered class diagrams
• Evaluation results

(Paste images here for documentation.)

9. Conclusion

The system successfully replicates the baseline methodology of the research paper. It demonstrates that local LLMs can generate UML class diagrams with reasonable syntactic and pragmatic quality, while semantic accuracy still requires improvement.

The integration of rule-based size metrics improves evaluation reliability.